Status	Finished
Started	Tuesday, 20 October 2026, 12:35 PM
Completed	Tuesday, 20 October 2026, 1:00 PM
Duration	25 mins 15 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Insertion

b. Complexity

c. Selection

d. Bubble ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list is sorted

b. When the list contains duplicate elements

c. When the element is found in the list ✓

d. When the search algorithm finishes

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

Number values should only be present

b. There should be direct access to the middle element in any sublist

c. The list must be sorted

d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It requires the list to be sorted ✓

b. It can only be applied to large lists

c. It is slower than linear search for small lists

d. It does not work with negative numbers

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Bubble search

b. Linear search ✓

c. Merge search

d. Quick search

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By comparing each element sequentially
- b. By dividing the list length by two ✓
- c. By using a hash function
- d. By starting from the first element

Question **7**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

- a. Hash search
- b. Sequential search
- c. Linear search
- d. Binary search ✓

Question **8**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O(1)
- b. O(log n)
- c. O(n log n)
- d. O👎 ✓

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search is more efficient than binary search on large lists.
- b. Linear search can be applied to both sorted and unsorted lists. ✓
- c. Linear search divides the list into halves.
- d. Linear search requires the list to be sorted.

Question **10**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search does not require dividing the list
- b. Binary search works on unsorted lists
- c. Binary search has a lower time complexity on large, sorted lists ✓
- d. Binary search can find multiple instances of the target element

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Binary search
- b. Linear search ✓
- c. Hashing
- d. Dynamic search

Question **12**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The lower middle element is chosen as the middle element ✗
- b. The middle element is chosen randomly
- c. The search stops
- d. The higher middle element is chosen as the middle element

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By dividing the list into halves
- b. By comparing each element in the list sequentially ✓
- c. By using a hash function
- d. By sorting the list first

Question **14**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- a. An error is raised
- b. The search is considered unsuccessful ✓
- c. The first element is returned
- d. The last element is returned

Question **15**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Binary search
- b. Linear search ✓
- c. Hash search
- d. Interpolation search